

WHARTON, TX, USA

WMO: 722351

Lat: 29.254N

Lon: 96.154W

Elev: 100

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12953

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.5	33.8	18.0	13.7	45.5	22.2	16.7	48.5	26.0	56.0	24.3	59.4	6.9	350	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.3	98.5	75.2	96.3	75.6	94.6	75.6	80.6	88.8	79.8	87.9	78.8	87.0	8.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.3	83.6	77.5	143.5	82.8	77.0	141.2	82.4	44.1	89.3	43.3	87.5	42.4	87.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	19.8	17.8	DB	25.9	102.2	3.6	2.5	23.4	104.0	21.3	105.5	19.3	107.0	16.7	108.8
			WB	24.4	81.0	3.8	1.2	21.6	81.8	19.4	82.5	17.3	83.1	14.5	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.2	53.0	57.9	63.9	69.8	76.5	82.1	83.4	84.5	80.5	71.8	62.5	55.7	(6)
	DBStd	12.94	9.15	9.75	8.17	6.46	4.98	2.86	2.70	3.11	4.01	7.39	9.32	9.49	(7)
	HDD50	181	72	36	8	1	0	0	0	0	0	0	15	49	(8)
	HDD65	1266	382	234	120	29	2	0	0	0	0	28	154	317	(9)
	CDD50	7557	167	257	439	596	823	962	1035	1070	915	678	390	225	(10)
	CDD65	3164	11	36	86	174	360	512	570	605	465	240	78	27	(11)
	CDH74	30944	40	127	422	1245	3223	5466	6276	6879	4408	2223	532	103	(12)
	CDH80	13823	0	6	42	270	1180	2645	3116	3624	2021	830	86	3	(13)
(14) Wind	WSAvg	7.0	8.0	8.8	8.8	9.0	7.9	6.0	5.2	4.8	4.9	5.9	6.8	7.6	(14)
(15) Precipitation	PrecAvg	43.7	3.2	2.6	2.0	2.4	4.4	4.9	3.3	3.3	6.0	4.7	3.0	3.0	(15)
	PrecMax	79.0	8.5	7.2	5.7	7.0	13.8	16.1	8.8	7.3	14.1	16.6	8.3	7.2	(16)
	PrecMin	24.7	0.2	0.3	0.2	0.0	0.4	0.3	0.3	0.2	0.9	0.2	0.1	0.6	(17)
	PrecStd	11.8	2.0	1.6	1.7	1.9	3.1	4.1	2.4	2.0	3.7	4.1	2.0	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.0	80.7	83.7	89.3	93.2	99.2	98.9	101.7	98.5	91.9	85.8	80.0	(19)
		MCWB	64.7	65.9	66.4	69.8	72.6	73.2	75.1	77.5	72.4	72.4	70.6	69.1	(20)
	2%	DB	74.6	77.7	81.1	85.9	90.7	95.4	96.8	99.2	95.0	90.0	82.3	77.1	(21)
		MCWB	64.2	66.7	67.7	69.1	72.9	75.2	75.6	76.1	74.7	73.2	70.2	68.8	(22)
	5%	DB	72.0	75.0	78.9	82.5	88.9	93.3	94.9	97.0	92.6	87.8	80.2	73.9	(23)
		MCWB	64.6	66.2	67.2	69.1	72.8	75.6	76.0	76.0	74.4	72.7	68.8	67.4	(24)
	10%	DB	69.1	72.4	76.5	81.0	86.6	91.1	92.7	94.9	90.4	85.5	77.1	71.7	(25)
		MCWB	63.0	65.8	66.3	68.7	72.9	75.5	76.1	76.3	74.9	71.3	68.1	66.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	71.1	72.3	75.7	78.0	80.6	81.3	82.0	81.1	79.4	75.2	73.2	(27)
		MCDB	72.0	75.2	78.7	82.1	85.2	88.2	89.6	91.0	88.4	85.7	80.9	76.1	(28)
	2%	WB	68.6	70.0	71.1	74.3	76.9	79.5	80.2	80.7	79.7	77.7	73.4	71.3	(29)
		MCDB	71.5	73.6	76.5	80.6	84.6	87.8	87.9	89.7	87.4	83.6	77.9	74.5	(30)
	5%	WB	66.6	69.1	70.0	73.0	76.0	78.4	79.2	79.8	78.4	75.9	72.0	69.7	(31)
		MCDB	70.9	72.7	74.6	79.1	83.6	86.6	86.5	88.9	85.8	82.7	77.0	72.8	(32)
	10%	WB	64.0	67.8	69.0	71.9	75.2	77.7	78.3	78.7	77.4	74.4	70.2	67.6	(33)
		MCDB	67.0	71.8	73.4	77.5	82.5	85.7	86.2	87.6	84.7	81.1	75.1	71.1	(34)
(35) Mean Daily Temperature Range	MDBR		20.3	19.4	19.5	19.7	18.4	18.5	18.8	20.3	19.2	22.5	21.0	19.4	(35)
	5% DB	MCDBR	22.2	19.7	19.0	20.6	20.2	21.7	22.1	23.3	22.2	22.3	21.2	20.6	(36)
		MCWBR	14.3	11.5	9.5	9.4	6.7	5.9	6.1	5.9	7.0	8.8	11.1	13.9	(37)
	5% WB	MCDBR	18.7	16.1	15.5	15.5	16.0	17.5	18.0	20.0	18.1	18.1	17.9	18.1	(38)
		MCWBR	14.8	11.8	9.4	8.5	6.1	6.0	6.2	6.3	7.0	9.0	11.3	14.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.344	0.356	0.377	0.431	0.483	0.457	0.485	0.472	0.447	0.383	0.375	0.346	(40)
	taud		2.482	2.457	2.389	2.244	2.134	2.267	2.204	2.242	2.277	2.444	2.411	2.485	(41)
	Ebn at Noon		283	289	289	275	260	266	258	260	263	276	269	276	(42)
	Edh at Noon		30	33	37	44	50	43	46	44	41	33	31	28	(43)
(44) All-Sky Solar Radiation	RadAvg		916	1047	1411	1685	1885	2012	1983	1889	1605	1388	1045	822	(44)
	RadStd		94	180	123	139	130	142	182	123	158	155	114	101	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+176

Nomenclature: See separate page